SUPPLEMENTARY MATERIALS: WHEN CAN MARKETS CRASH WITHOUT NEWS?

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SM1. Scope of the supplement. These panels and the earlier nonoptimal two-day witness are preserved for replication and illustration. They are not independent estimates of real market risk. All mathematical claims and the improved interval-certified construction appear in the main manuscript.

SM2. Earlier exploratory two-day execution. The preceding theorems form a hierarchy of *necessary* tests. Passing the rolling-band bound, the scalar selling-rate test, the dollar-closure preconditioning test, and the cut-capacity test still does not prove a path exists. We therefore turn from exclusion and lower bounds to a constructive example that enforces the constraints *simultaneously*. To test repeated *partial-coverage* lower-limit closes, we constructed a continuous $N = 10$, $k = 9$ example in the $g(S) \equiv \eta$ market. The nine targets use symmetric $\pm 3\%$ proportional bands; the compensator uses a wider symmetric $\pm 25\%$ band. These are *hypothetical normalized assets*, not calibrated TSE stocks. Set $F = \eta = 1$, $\rho = 0.4$, equal session lengths $\tau_1 = \tau_2 = 20$, and a common gross share-rate limit $\bar{v} = 1$.

Write $S_1 = \cdots = S_9 = x(t)$ and $S_{10} = y(t)$. On each session let $x(t)$ be a continuous piecewise-linear path: it rises to a first-session plateau h_1 then closes at 0.97, rises to a second-session plateau $h_2 = 0.995$ then closes at $0.97^2 = 0.9409$. All ramps last 0.7 trading-time units. Dollar pairing uniquely determines the compensator's ODE via

$$u(t) = x(t)[\dot{x}(t) + \rho(x(t) - 1)], \qquad \dot{y} = -\rho(y - 1) - \frac{9u(t)}{y(t)}, \quad y(0) = 1. \tag{SM2.1}$$

The first plateau $h_1 \approx 1.0142982125$ is selected from the *exact piecewise-polynomial* equation

$$X_i(T_2) = \frac{x(T_2)^2 - 1}{2} + \rho \int_0^{T_2} x(t)[x(t) - 1]\,\mathrm{d}t = 0, \quad i = 1, \ldots, 9. \tag{SM2.2}$$

Because $\dot{X}_{10} = -9\dot{X}_1$, the compensator's terminal executed-dollar balance closes as well. Independent high-accuracy ODE and Runge–Kutta integrations agree for $y(T_2)$ to approximately 2×10^{-12}; band and rate slack were checked on a dense time grid. These numerical checks are an explicit *witness*, not a proof of general N-asset sufficiency or a certified interval-enclosure theorem.

The example proves neither a trader's incentive to undertake the sequence nor a *locked* limit-down order book. It does show numerically that rolling bands, finite executed-share rates, no fundamental-news shocks, and terminal dollar closure can be compatible with *multiple* lower-limit closes in a broad subset of assets. For a crisis threshold, the breadth requirement alone does not suffice: for this path $Q_{0.9}(T_2) = 0.060918$ but the average-loss restriction binds, giving $\mathcal{C}_{0.9}(T_2) = 0.019932$ (thus, for example, $\delta = 0.015$ is reached). No claim of an exact minimum rate, minimum turnover, or equilibrium occurrence follows.

SM3. Other diagnostic displays.

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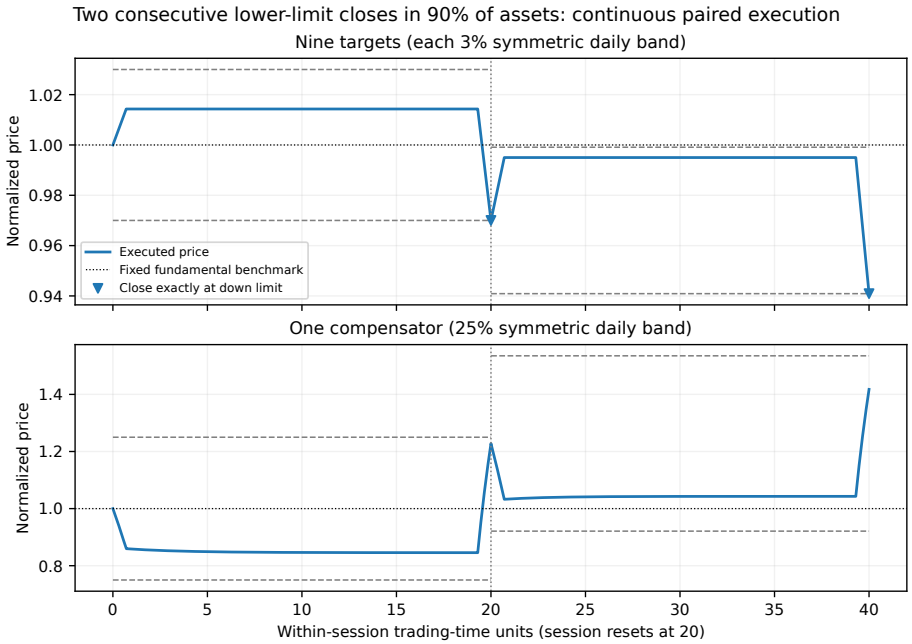


FIG. SM1. ***Repeated partial-coverage lower-limit closing prices under rolling daily bands.*** *The same nine target assets close at their 3% lower limits twice. The tenth, higher-band compensating asset stays within its own 25% limits. The displayed paths contain no instantaneous trades. The horizon and bands are specified model inputs, not observations.*

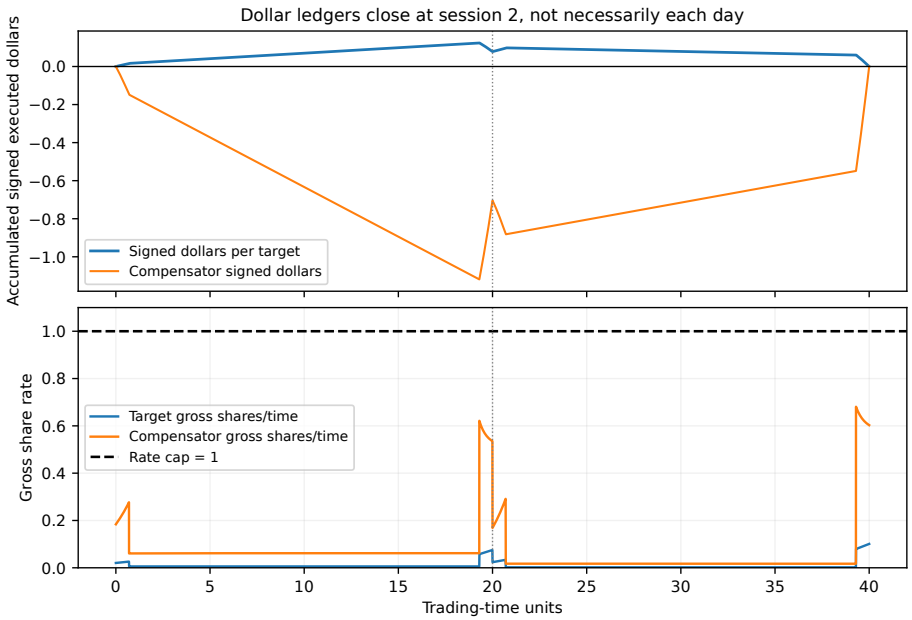


FIG. SM2. ***Trading ledgers and gross-rate check for the same two-session witness.*** *The first session builds positive net dollar flow in the targets; the second reverses it, leaving every asset's final signed-dollar balance zero. Gross rates remain below the imposed share-rate limit; dollar balance is enforced continuously, not just at maturity.*

TABLE SM1

Two-session $N = 10$ deterministic closing-limit witness (normalized units). Day 1 and Day 2 target losses are 3% each relative to the reset reference; the terminal systemic functional is smaller because the compensator rises.

Quantity	Verified value
Nine target closing prices, session 1 / session 2	0.970000 / 0.940900
Compensator closing prices, session 1 / session 2	1.228104 / 1.417574
Maximum individual gross executed-share rate	$0.680780 < \bar{v} = 1$
Target executed-dollar balance, day 1 / day 2	$+0.078235$ / -0.078235
Maximum absolute terminal net-dollar balance	$< 7 \times 10^{-15}$
Total paired-dollar turnover / largest gross share sales	2.59310 / 1.46241
Terminal $\mathcal{C}_{0.9}(T_2)$ / breadth loss $Q_{0.9}(T_2)$	0.019932 / 0.060918

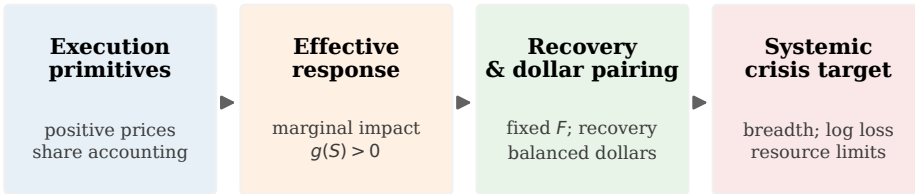


FIG. SM3. ***One explicitly defined ideal market.*** *Price priority, signed-share accounting, effective price impact, resilience, and paired-dollar trading have distinct mathematical roles. All quantitative panels in this paper are illustrations of this specified model rather than observations of a live order book.*

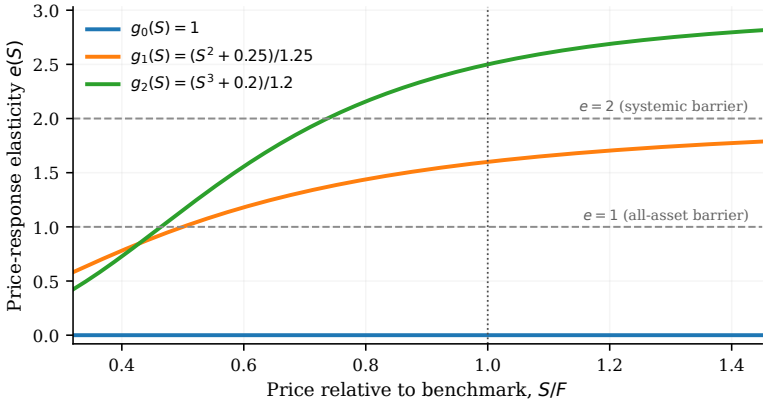


FIG. SM4. ***Local elasticity of three positive, smooth, non-power-law impact functions.*** *The dotted vertical line marks the common initial price $S = F$. The global elasticity-barrier theorems in the main manuscript require their elasticity inequalities throughout the* relevant attainable price region*; a value at F alone gives no global classification. This is a formula-based illustration, not empirical market depth.*

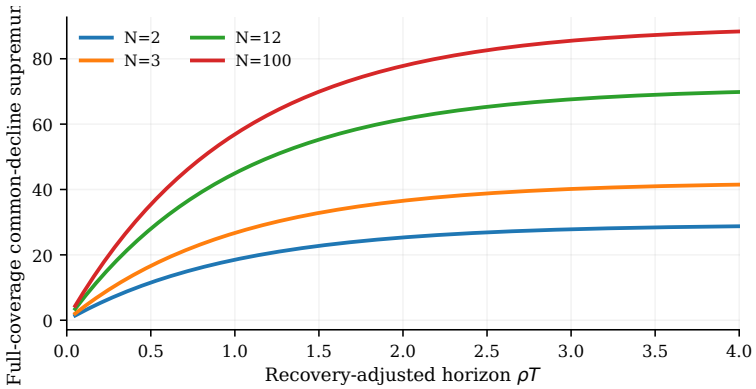


FIG. SM5. ***A sharp, model-specific common-decline boundary.*** *For $g = \eta$ and $k = N$, the upper envelope of the common simple-price decline is $100(1 - e^{-\delta_{\mathrm{crit}}})\%$. Every strictly smaller log-loss threshold is reachable with some finite budget; the boundary itself is not attained with strictly positive prices. This is not a probability curve.*

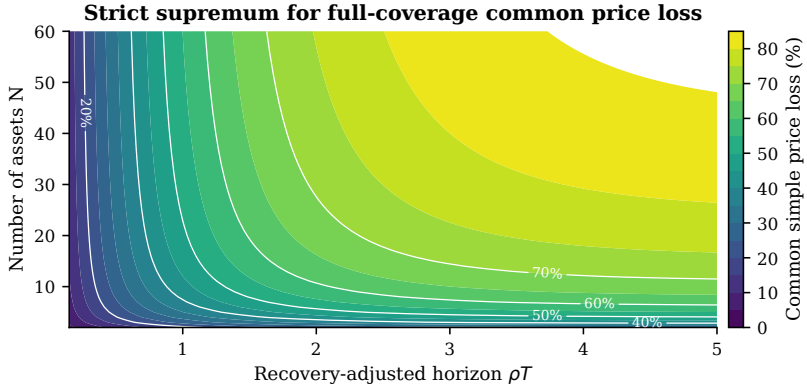


FIG. SM6. ***The full-coverage boundary across asset count and recovery time.*** *Color reports* $100(1-e^{-\delta_{\mathrm{crit}}(N,T)})$*, the supremum of simultaneous* simple-price *losses for constant per-share impact with individually closed terminal dollar flows. No probability, observed drawdown, or finite-budget optimum is inferred.*

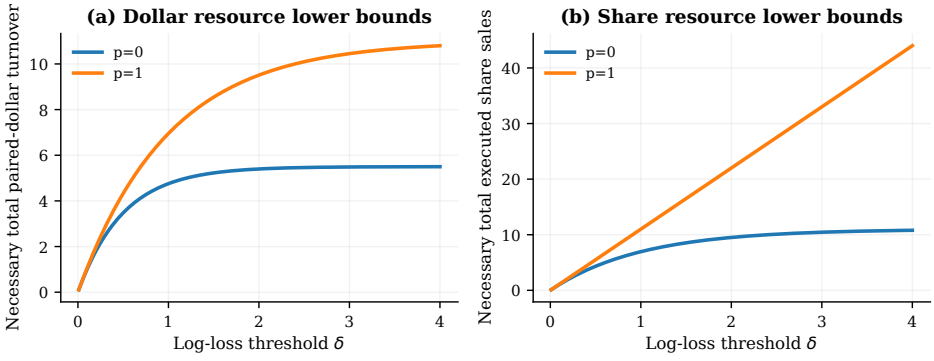


FIG. SM7. ***General necessary resource floors illustrated for the power-response family.*** *With* $F=\eta=1$ *and* $k=11$*, the curves evaluate* $k\int_{e^{-\delta}}^{1} S/g(S)\,\mathrm{d}S$ *(dollars) and* $k\int_{e^{-\delta}}^{1} 1/g(S)\,\mathrm{d}S$ *(shares), for* $p=0,1$*. The bounds are* not *the optimal execution cost and do not establish reachability.*

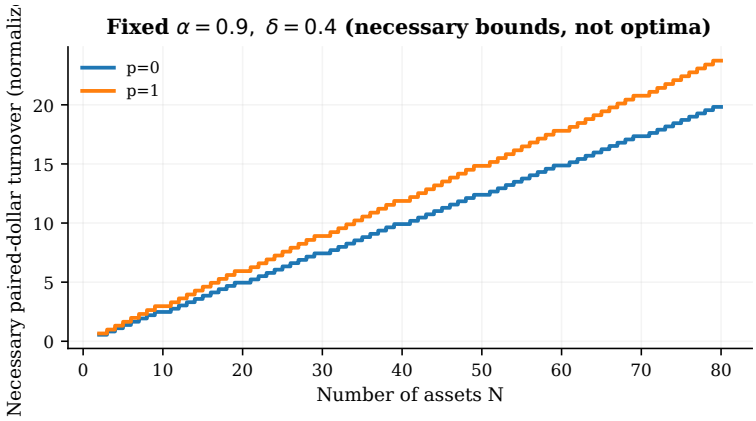


FIG. SM8. ***Dependence on asset number of the necessary dollar-resource bound.*** *The experiment copies per-asset benchmark price and response parameters, fixes* $\alpha = 0.9$ *and* $\delta = 0.4$*, and increases the number of assets* N*. The steps arise from the discrete coverage count* $\lceil \alpha N \rceil$*. It is not a capitalization-scaled prediction for an actual stock index.*

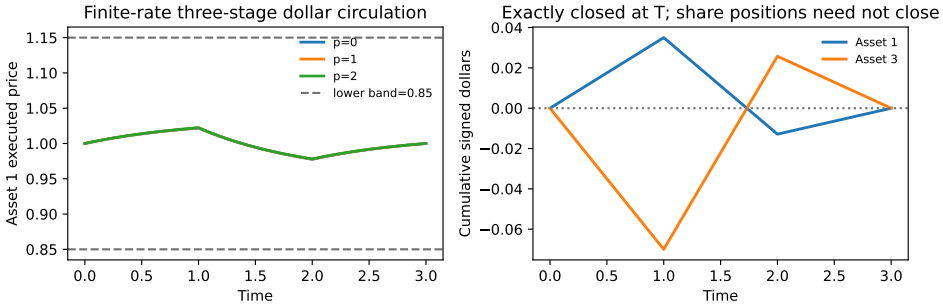


FIG. SM9. ***A new bounded-rate, terminally dollar-closed path (not a pulse approximation).*** *For* $N = 3$*,* $T = 3$*,* $a = .035$*,* $F = \rho = \eta = 1$*, and* $u = (1, 1, -2)$*, the price trajectories under* $p = 0, 1, 2$ *share the same absolutely continuous dollar-rate signal; right panel shows dollar ledgers that close exactly. The peak per-asset gross share rate is below* 0.15*, and all prices remain inside a* $\pm 15\%$ *band.*

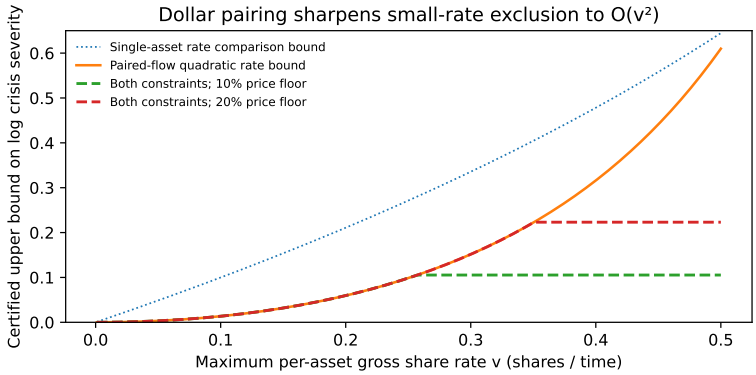


FIG. SM10. ***Exclusion certificates, not exact reachable maxima.*** *For* $g = \eta = F = \rho = 1$ *and* $T = 3$*, a paired-flow quadratic estimate improves the single-price comparison at small rates; a single-session lower price band provides an additional hard ceiling. Neither curve reports optimal crisis control or an observed market frequency.*

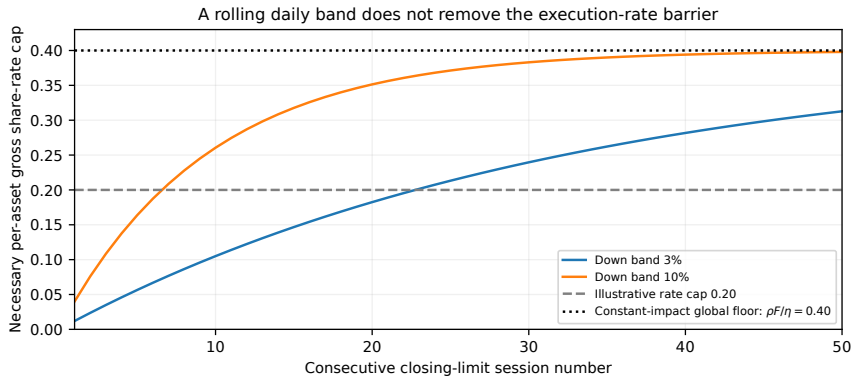


FIG. SM11. ***Necessary gross share-rate floor for repeated lower-limit closes.*** *The scalar constant-impact necessary-rate formula in the main manuscript in the scalar constant-impact comparator,* $F = \eta = 1$, $\rho = 0.4$, $\tau = 20$. *These are* necessary *per-asset rates; paired execution, available inventory, and the compensator's own daily band can make the actual market threshold higher.*

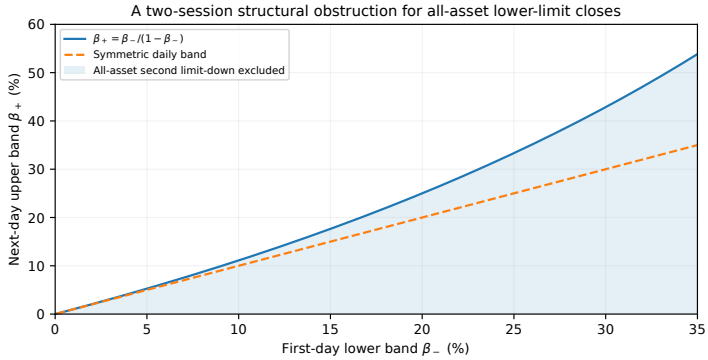


FIG. SM12. ***Parameter region in which the second all-asset lower-limit close is ruled out.*** *The region beneath the curve satisfies* $(1+\beta_+)(1-\beta_-) \le 1$. *Symmetric proportional bands lie strictly inside this region. This is a conditional invariant-region result, not a model of exchange-wide order queues.*

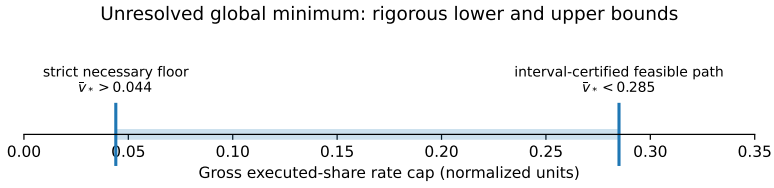


FIG. SM13. ***What is and is not optimized.*** *A strict rate floor from the existing quadratic systemic bound and a certified feasible execution path bracket the unknown global minimum* $\bar{v}_*$. *The interval is a logical admissibility bracket, not a statistical confidence interval.*

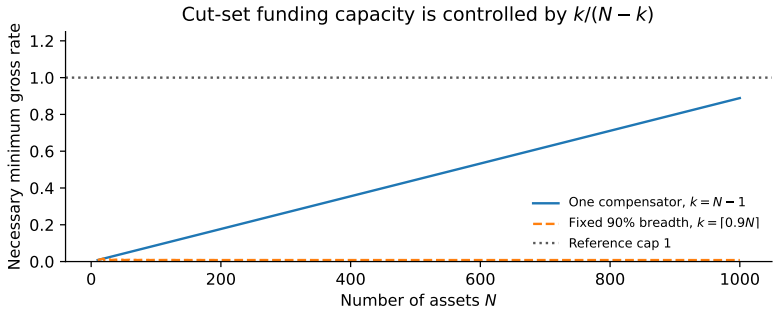


FIG. SM14. ***A funding-capacity constraint that grows with the falling majority.*** *The necessary second-day rate in the main manuscript increases as $k/(N-k)$ when a single asset absorbs the net matched dollars of an expanding majority. This does not prove that a feasible market exists at every plotted N.*

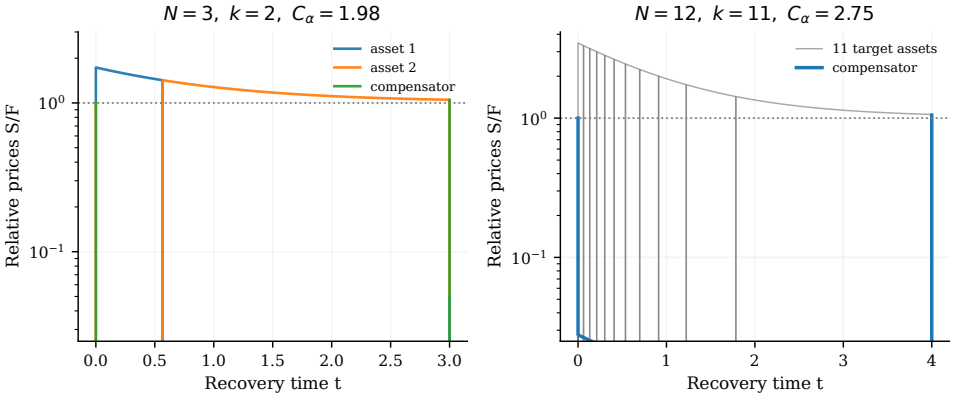


FIG. SM15. ***Constructed price paths with terminally closed dollar flows.*** *For $N = 3$ and $N = 12$, one compensating asset ends above its initial value, while the other $N - 1$ prices fall by the common target log loss $L = 3$. Logarithmic vertical scales keep near-zero prices visible. Kinks and jumps are permitted control pulses, not simulated exchange events.*

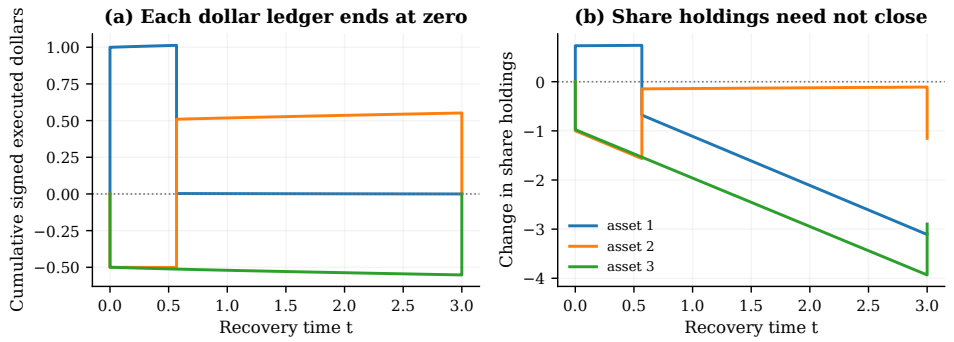


FIG. SM16. ***Net executed dollars are not net shares.*** *A separate $N = 3$, $k = 2$, $T = 3$, $L = 5$ witness has $X_i(T) = 0$ for every asset (left), but its net share positions generally differ from zero (right). Initial share inventories must cover gross sales; transaction closure alone does not supply them.*

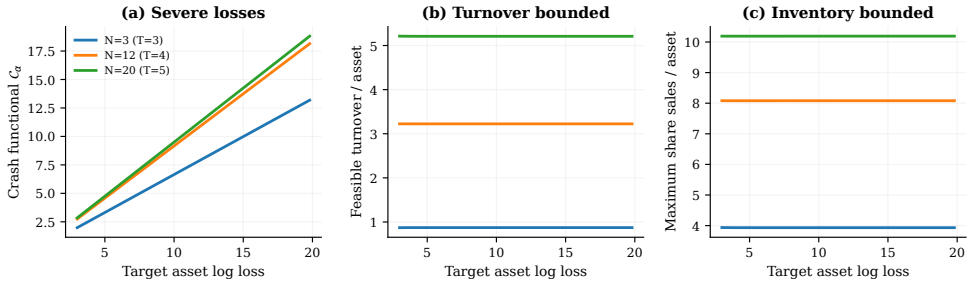


FIG. SM17. ***Increasing crisis severity with bounded* feasible *resources.*** *Selected* $N = 3, 12, 20$ *construction families use* $k = N-1$ *and horizons* $T = 3, 4, 5$*, respectively. As the terminal target-price log loss increases, achieved* $\mathcal{C}_\alpha$ *rises (left), whereas the particular paths' turnover per asset (center) and largest cumulative share sale (right) remain bounded. The different horizons and prices are model choices, not a cross-market calibration or an optimum.*

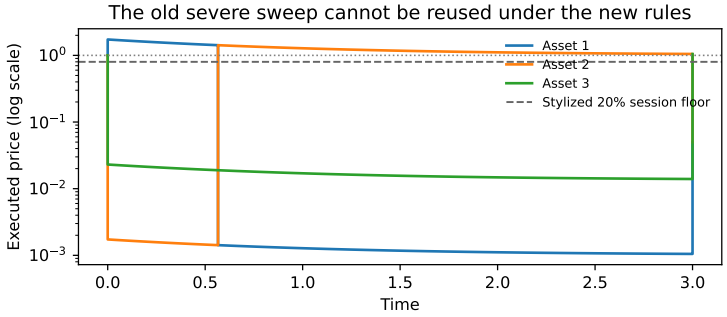


FIG. SM18. ***Diagnostic rejection of an old severe path under new execution restrictions.*** *The* $N = 3$*,* $p = 0$ *severe-dollar-closed witness from the unrestricted model crosses a stylized one-session 20% lower band and includes nonzero instantaneous pulses. Its exact closed dollar ledger therefore* does not *certify reachability under either a finite gross rate or the price-limit protocol.*